DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture Week 6: Assignment

Multi-Period Trees · Martingales · American Options

Instructions

- This assignment covers Lectures 16–18 and the Week 6 Lab. It carries **100 marks** total.
- Show all working for numerical questions – correct answers with no working receive partial credit only.
- Questions marked **[Python]** require a short script; submit your code alongside your written answers.
- Submit a single PDF (typeset or clearly scanned handwriting) plus any `.py`/`.ipynb` files, via the course submission portal.

Part A – Concepts (15 marks)

A1. **(4 marks)** Explain why the binomial lattice *recombines*, and quantify the computational consequence: how many nodes does a 40-step recombining tree have, versus how many end nodes a non-recombining tree would have?

A2. **(4 marks)** State the definition of a discrete-time martingale, and explain in one or two sentences why it is described as a "fair game."

A3. **(4 marks)** State the First Fundamental Theorem of Asset Pricing, and explain what *market completeness* adds to it.

A4. **(3 marks)** True or false, with justification: "Backward induction is a computational shortcut with no deeper theoretical justification."

Part B – Multi-Period Trees (25 marks)

Use the following model for questions B1–B3: $S_0 = 60$, $u = 1.15$, $d = 0.9$, $r = 4\%$ per step, $N = 3$ steps.

B1. **(6 marks)** Compute and tabulate the full stock lattice $S_{n,j}$ for $n = 0, 1, 2, 3$. State how many nodes there are at each time step.

B2. **(4 marks)** Compute the risk-neutral probability q for this model, and verify the no-arbitrage condition $d < 1 + r < u$ holds.

B3. **(10 marks)** Price a European call with strike $K = 62$ by backward induction, showing the option value at *every* node of the tree (not just the final answer).

B4. **(5 marks)** Price the corresponding European put with $K = 62$, then verify put–call parity numerically.

Part C – Martingales (20 marks)

C1. (6 marks) For $S_0 = 80$, $u = 1.25$, $d = 0.8$, $r = 5\%$: compute q, then verify explicitly that $\mathbb{E}^Q[\tilde{S}_1 \mid \mathcal{F}_0] = \tilde{S}_0$, where $\tilde{S}_n = S_n/(1+r)^n$.

C2. (5 marks) Using the same model but a real-world probability $p = 0.65$, compute $\mathbb{E}^P[\tilde{S}_1]$ and compare it to $\tilde{S}_0$. State the sign of the difference and give its economic interpretation.

C3. (5 marks) Starting from the martingale property $\mathbb{E}^Q[\tilde{V}_{n+1} \mid \mathcal{F}_n] = \tilde{V}_n$ for the discounted option value, derive the backward induction recursion $V_n = \frac{1}{1+r}[qV_{n+1,\text{up}} + (1-q)V_{n+1,\text{down}}]$. Show each algebraic step.

C4. (4 marks) A gambler wins Rs. 1 with probability 0.48 and loses Rs. 1 with probability 0.52 on each round. Is their wealth a martingale, submartingale, or supermartingale? Justify your answer using the definition.

Part D – American Options (25 marks)

D1. (12 marks) Using the Part B model ($S_0 = 60$, $u = 1.15$, $d = 0.9$, $r = 4\%$, $N = 3$, $K = 62$), price the **American** put by modified backward induction. At every node, report both the continuation value and the exercise value, and state clearly which nodes (if any) call for early exercise. Compute the early exercise premium relative to your B4 answer.

D2. (5 marks) Explain, in your own words, why an American *call* on a non-dividend-paying stock should never be exercised early. Give both of the reasons discussed in Lecture 18.

D3. (4 marks) Explain why the compact one-shot formula $V_0 = (1+r)^{-N}\mathbb{E}^Q[\text{payoff}]$ is valid for European options but *fails* for American options.

D4. (4 marks) Explain why put–call parity, as stated in Lecture 18, holds for European options but not for American ones. What structural feature of the derivation breaks down?

Part E – Computational (10 marks)

E1. (10 marks) [Python] Using your Lab code, take $S_0 = 100$, $K = 100$, $r = 5\%$ p.a., $\sigma = 30\%$, $T = 1$ year, and the CRR parameterization. Price the **American put** for $N = 10, 25, 50, 100, 250, 500$ steps, and plot the price against N. Comment in 2–3 sentences: does the American put price converge? Does it converge from above or below, and is the convergence smoother or more oscillatory than the European call convergence you saw in the Lab?

Part F – Integrative Question (5 marks)

F1. (5 marks) In Week 5 you priced a one-period option two ways: by building a replicating portfolio, and by risk-neutral valuation. In Week 6 you learned that discounted prices are martingales under Q. In a short paragraph (100–150 words), explain how these three ideas – replication, risk-neutral valuation, and the martingale property – are really three views of the *same* underlying fact, and identify which of the three the First Fundamental Theorem of Asset Pricing generalizes to arbitrary markets.

End of Assignment. Solutions will be released after the submission deadline.